

1.Introduction

The demand for automated systems in agriculture and retail continues to grow. Hence, our team is developing a Vegetable Image Classifier using convolutional neural networks (CNNs). The goal is to create a model that can classify vegetables. While this is a small scope work, the project can be extended to many objects and other sectors in the real world.

This work is built from scratch for the purpose of education and learning.

2.Motivation and Impact

This project is a hands-on opportunity to explore machine learning techniques.

Who Benefits?

- Farmers: Automate sorting and quality checks.
- Retailers: Support for self-checkout and inventory systems.
- Consumers: Faster, more accurate checkout experience.
- Developers: Potential for integration in smart grocery platforms like Amazon Go.

Dataset and Setup

We use the [Vegetable Image Dataset](#) from Kaggle, which contains 21,000 images of 15 vegetable types. Each class has 1,400 images (224×224 resolution, JPG format). It has 15,000 images for training, 3,000 images for validation, and 3,000 images for testing.

The dataset includes real-world variability in lighting and background, making it suitable for training robust models.

Model and Evaluation

We use a supervised CNN model with 3–4 convolutional layers and 2–3 dense layers, aiming for a balance between simplicity and accuracy.

Evaluation metrics include: Accuracy, Confusion Matrix, Precision, Recall, F1 Score, R^2 and Variance, Visualization (loss/accuracy curves, error analysis).

As a baseline, we reference ResNet-50, a widely used deep learning architecture, to evaluate how our custom model compares to a standard benchmark.

Risks and Mitigation

Like all ML projects, this one faces common challenges:

- Overfitting: when Model memorizes data.
 - *Mitigation*: Use data augmentation, regularization, and avoid overly complex models.
 - Underfitting: when the model too simple to capture patterns.
 - *Mitigation*: Adjust architecture based on validation feedback.
 - Poor preprocessing: as Noise or imbalance can reduce accuracy.
 - *Mitigation*: Clean, well-balanced dataset with minimal, effective preprocessing.
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Conclusion

The Vegetables Identifier will be a good milestone for learning about machine learning and has practical applications. With a good dataset, structure and plan, the project has the potential to contribute to the agriculture and retail industry. We start a flexible and extendable project as the baseline for future work.